



National University

Of Computer & Emerging Sciences Faisalabad - Chiniot Campus

CS 1005 : Discrete Structures

Assignment 4

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Section BCS (C)

Q No 01: An urn contains 5 blue and 7 gray balls. Let us say that 2 are chosen at random, one after the other, without replacement. [10]

- i. Find the following probabilities and illustrate them with a tree diagram: the probability that both balls are blue, the probability that the first ball is blue and the second is not blue, the probability that the first ball is not blue and the second ball is blue, and the probability that neither ball is blue.
- ii. What is the probability that the second ball is blue?
- iii. What is the probability that at least one of the balls is blue?
- iv. If the experiment of choosing two balls from the urn were repeated many times over, what would be the expected value of the number of blue balls?

Q No 02: Suppose that one urn contains 3 blue and 4 gray balls and a second urn contains 5 blue and 3 gray balls. A ball is selected by choosing one of the urns at random and then picking a ball at random from that urn. If the chosen ball is blue, what is the probability that it came from the first urn? [10]

Q No 03: Consider a medical test that screens for a disease found in 5 people in 1,000. Suppose that the false positive rate is 3% and the false negative rate is 1%. Then 99% of the time a person who has the condition tests positive for it, and 97% of the time a person who does not have the condition tests negative for it. (See exercise 4 at the end of this section. [10]

- i. What is the probability that a randomly chosen person who tests positive for the disease actually has the disease?
- ii. What is the probability that a randomly chosen person who tests negative for the disease does not indeed have the disease?

Q No 04: Suppose A and B are independent events in a sample space S. Show that A and B^c are also independent. [5]

Q No 05: A coin is loaded so that the probability of heads is 0.6. Suppose the coin is tossed twice. Although the probability of heads is greater than the probability of tails, there is no reason to believe that whether the coin lands heads or tails on one toss will affect whether it lands heads or tails on the other toss. Thus it is reasonable to assume that the results of the tosses are independent. [10]



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- i. What is the probability of obtaining two heads?
- ii. What is the probability of obtaining one head?
- iii. What is the probability of obtaining no heads?
- iv. What is the probability of obtaining at least one head?

Q No 06: A coin is loaded so that the probability of heads is 0.6 (and thus the probability of tails is 0.4). Suppose the coin is tossed ten times that the results of the tosses are mutually independent. [5]

- i. What is the probability of obtaining eight heads?
- ii. What is the probability of obtaining at least eight heads?

Q No 07: A pool of 10 semifinalists for a job consists of 7 men and 3 women. Because all are considered equally qualified, the names of two of the semifinalists are drawn, one after the other, at random, to become finalists for the job. [10]

- i. What is the probability that both finalists are women?
- ii. What is the probability that both finalists are men?
- iii. What is the probability that one finalist is a woman and the other is a man?

Q No 08: One urn contains 10 red balls and 25 green balls, and a second urn contains 22 red balls and 15 green balls. A ball is chosen as follows: First an urn is selected by tossing a loaded coin with probability 0.4 of landing heads up and probability 0.6 of landing tails up. If the coin lands heads up, the first urn is chosen; otherwise, the second urn is chosen. Then a ball is picked at random from the chosen urn. [10]

- i. What is the probability that the chosen ball is green?
- ii. If the chosen ball is green, what is the probability that it was picked from the first urn?

Q No 09: A coin is loaded so that the probability of heads is 0.7 and the probability of tails is 0.3. Suppose that the coin is tossed twice and that the results of the tosses are independent. [10]

- i. What is the probability of obtaining exactly two heads?
- ii. What is the probability of obtaining exactly one head?
- iii. What is the probability of obtaining no heads?
- iv. What is the probability of obtaining at least one head?